Generative AI for Cosmological Simulation and Manifold Representation

Using Stable Diffusion Architecture

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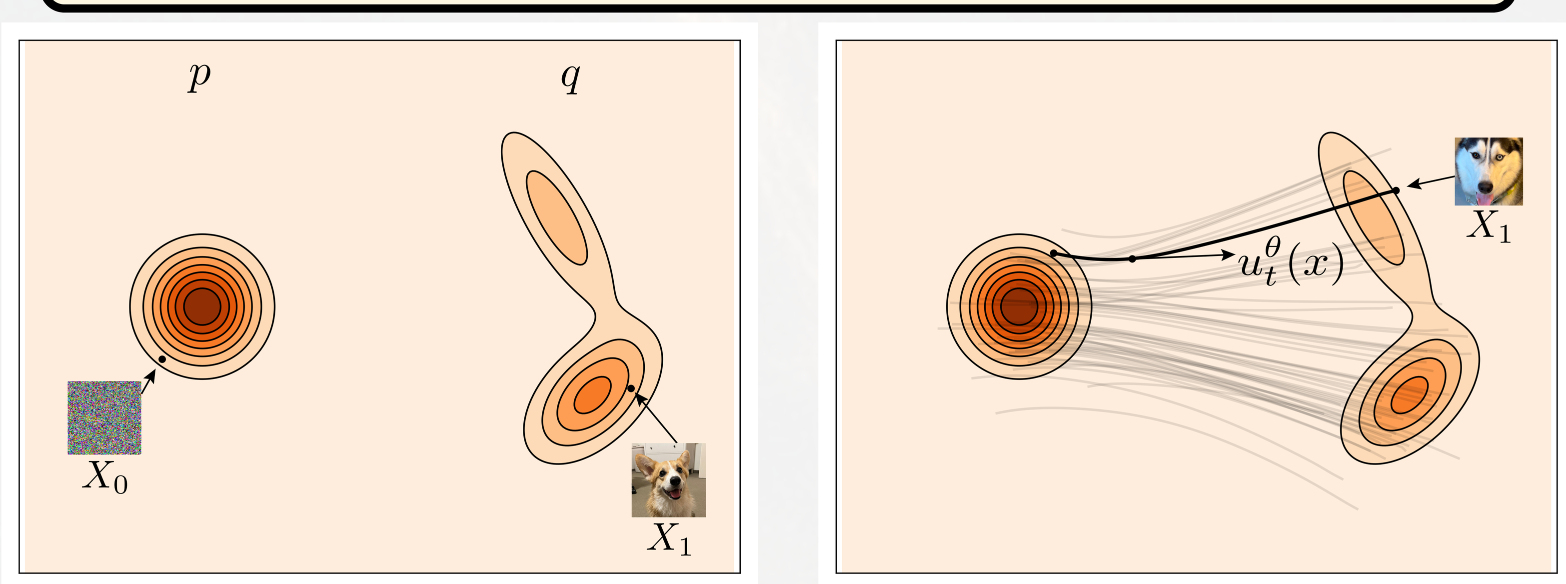
Introduction

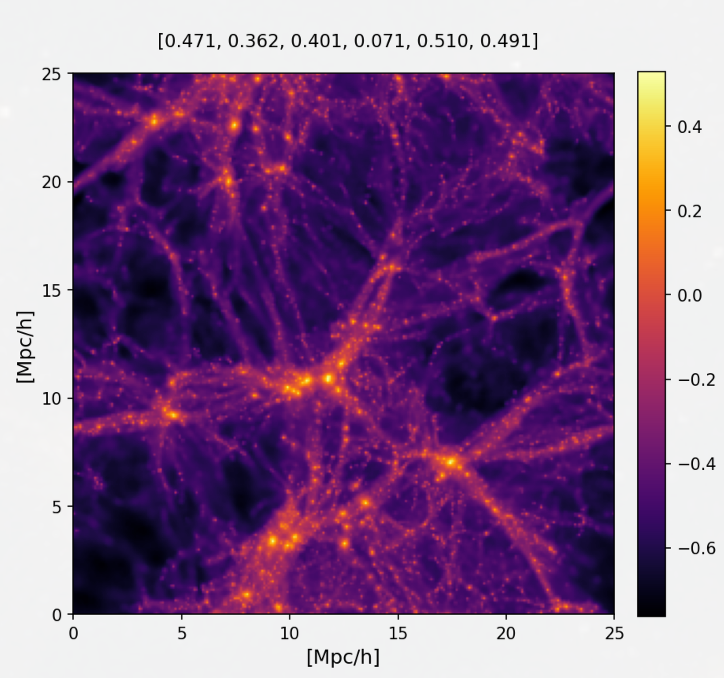
- Cosmological simulations are computationally expensive
- Recently **Generative AI** has emerged as a fast emulator capable of directly generating complex data $\boldsymbol{x_0}$ from conditioned on $\boldsymbol{c}$
- Worked on developing *Diffusion Model* to the CAMEL dataset to ***generate conditional cosmic maps*** that accurately reproduce key physical statistics.
- Furthermore, we analyze how the AI learns the nature of the low-dimensional *data manifold* defined by the 6 parameters $\boldsymbol{c}$

Generative AI



Generative AI synthesize data by learning the underlying probability distribution of the data and sampling from it. *Diffusion* and *Flow Matching* approaches achieve this by progressively transforming a simple Gaussian prior $\boldsymbol{p}$ into the target distribution $\boldsymbol{q}$.

Training Data



Total : *15000* $(\boldsymbol{x_0}, \boldsymbol{c})$

Cosmic 2D Image : $x_0 \in \mathcal{R}^{256\times256}$

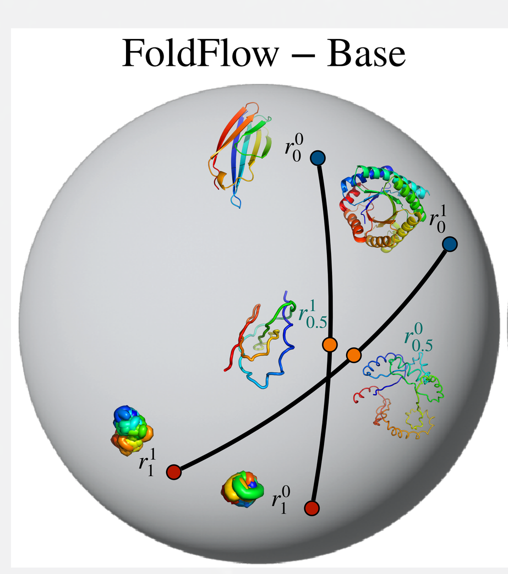
Cosmological Parameter : $c \in \mathcal{R}^6$

$\boldsymbol{\Omega_m}$	: total matter density
$\boldsymbol{\sigma_8}$.	: Amplitude of Matter Fluctuations
$\boldsymbol{A_{SN1}}$, $\boldsymbol{A_{SN2}}$	: Supernova Feedback Parameter (Wind Velocity, Mass Loading Factor)
$\boldsymbol{A_{AGN1}}$, $\boldsymbol{A_{AGN2}}$	: AGN Feedback Parameter (Energy Scale/Threshold, Energy Efficiency)

Used *IllustrisTNG* from the **CAMEL (Cosmology and Astrophysics with MachinE Learning Simulations)** dataset. It is a hydrodynamic simulation based on the AREPO code, which tracks cosmic evolution from early universe density fluctuations (z=127) down to the present day (z=0), computing both the gravitational interactions of dark matter and complex **Baryon Physics**.

Manifold Hypothesis

- The manifold hypothesis suggests that high-dimensional data actually lie on a low-dimensional manifold.
- This underlying probability distribution is often referred to as the *data manifold*.
- Generative AI models can thus be viewed as *manifold runners*, learning to navigate and reproduce the structure of this hidden space.



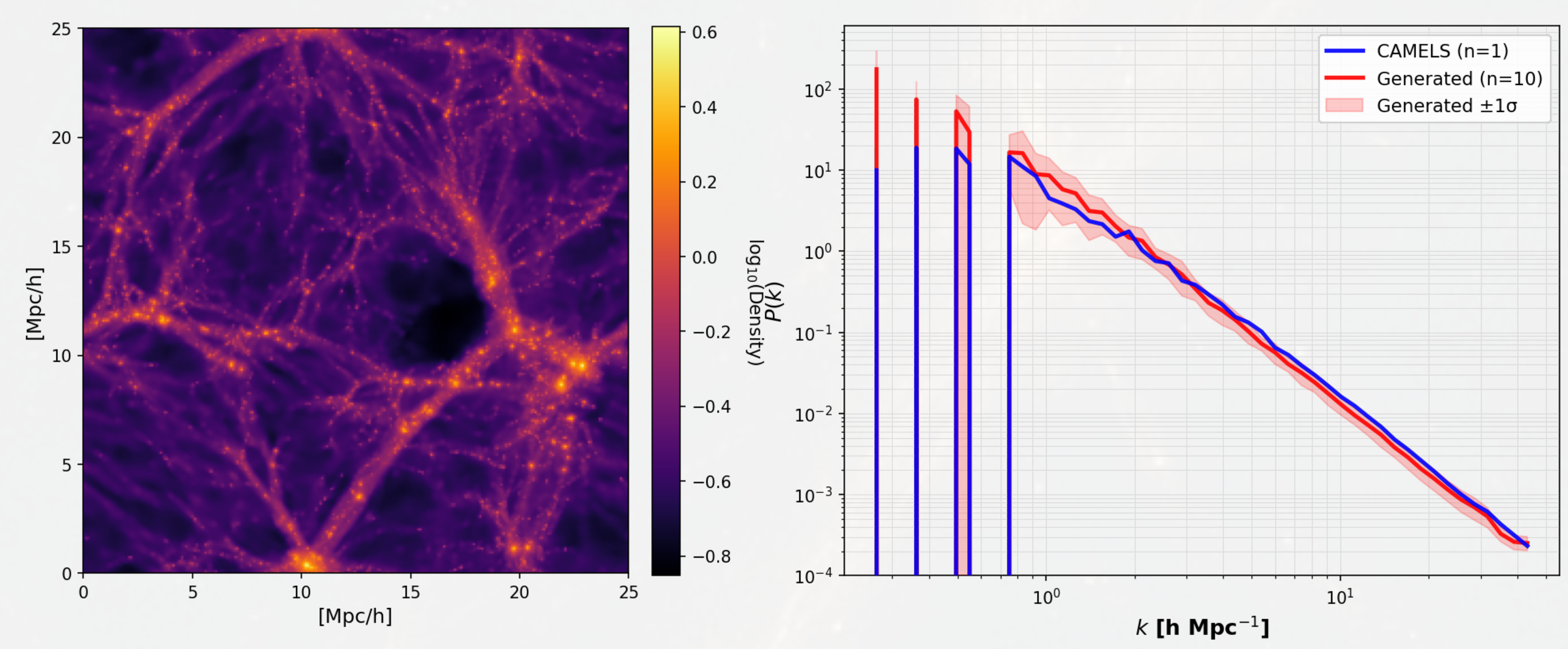
Conclusion & Out Look

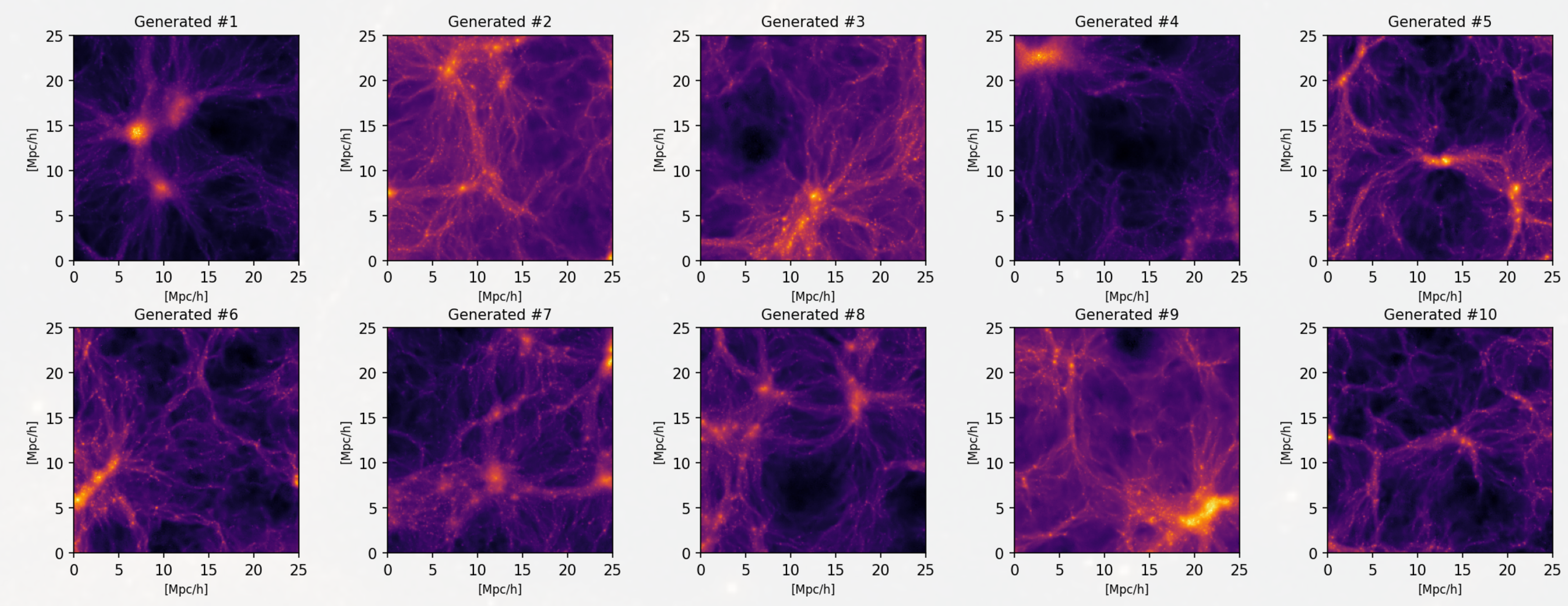
- Developed a generative model conditioned on cosmological parameters
- The generated halos and power spectra show generally good agreement with simulations, though some discrepancies remain.
- Explored whether the model has learned the underlying data manifold — results are still preliminary, but promising.

Power Spectrum Analysis

To test whether the model is ***cosmologically*** well trained, compared to the power spectrum.

$$\boldsymbol{c} = (\Omega_m, \sigma_8, A_{SN1}, A_{SN2}, A_{AGN1}, A_{AGN2})$$





Halo Mass Analysis

These plots compare the **halo mass function**s between real and generated density fields.

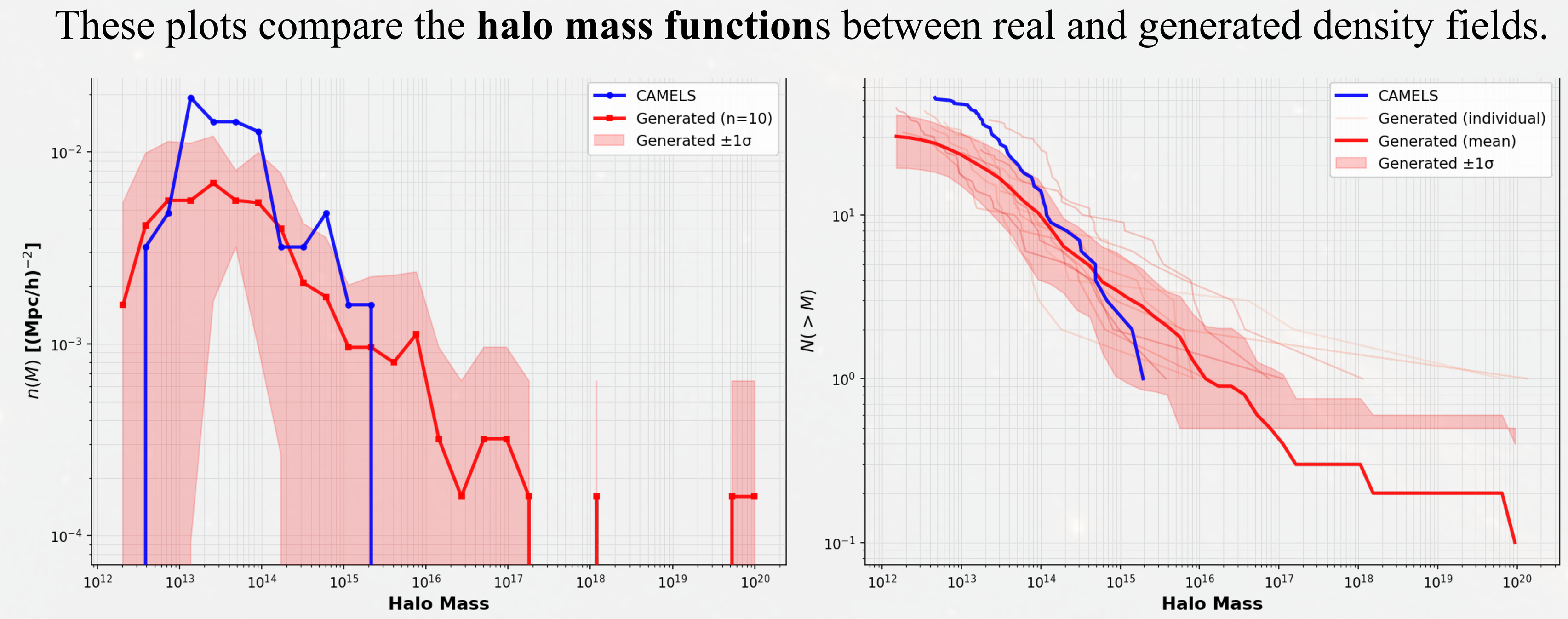


Figure demonstrates that the generative model reproduces the correct cosmological halo abundance and cumulative distribution, confirming its physical consistency beyond the power spectrum.

Does Diffusion Model Learned the Cosmic Manifold?